

Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

In this engineering design project our goal is to design a sturdy, protective, secure camera lens cover that protects the camera lens, stays secure to the camera (in on position), and doesn't get lost with string (in off position), with quantifiable constraints such as being 3.52 in. long, 3.01in in height, and weighing under 3 pounds.

Photographers and others using film equipment are involved in this problem. The problem itself is getting fingerprints, scratches, or other damage onto the camera lens. These lenses can cost hundreds of dollars, even up to the low thousands, so these can be very expensive to replace (Stoker, Gavin). Of course, fingerprints can be cleaned off, but scratches and chips cause permanent damage (Dowell). This hurts the quality of the picture. The people that we are trying to solve the problem for are the TV and filming crew at our high school. They have lost multiple lens covers, and require more.

The camera lens cover that we are going to produce will protect the camera lens when it is not in use. It simply fits over the lens covering, stopping any fingerprints when it is picked up, and protecting it from any damage when it is dropped or otherwise put in harm's way.

This problem is unique to camera lenses when they are left uncovered. When this happens, they are more vulnerable to being damaged by accidental scratches or dust particles. Because of this, there needs to be a product to cover the lens while it is not in use (Kentfaith). The school's television studio is currently missing some of their lens covers, and the covers that we will be creating are made to replace those.

This is a medium size issue, as lens scratches occur somewhat commonly to the camera, and this requires the entire camera to be replaced, but cameras aren't the most important object in the world to protect right now. The biggest aspect to the problem is that the image would see a drop in quality, but the image is still mostly legible otherwise. The school's cameras don't get damaged often, so it is not a pressing issue.

This problem could happen in the TV studio, or when filming out in the field. It really could happen anywhere. All that needs to happen for the problem to arise is for the camera to be dropped without a lens cover on.

The problem occurs whenever the camera is dropped without a lens cover on, or when people pickup the camera while grabbing the lens (Paul, Winchell).

The problem starts when the camera lens is left uncovered, giving it no protection against scratches. When the camera is left like this, it is possible for the

lens to be scratched by accident. These scratches can happen by accident, but dust landing on the lens can also damage it. (Stoker, Gavin)

Most camera lenses are scratch resistant, so scratches don't happen too often, however they are not scratch proof. With the lack of camera lenses at the school TV department, the cameras are at a much higher risk.

Stakeholder Concerns and Connections (A3, A4)

Walpole TV Studio, Walpole High sports teams, Walpole High students, news broadcasters, and photographers are in need of our lens covers. These stakeholders would suffer a great magnitude of monetary losses if they do not receive our lens cover. If the Walpole TV Studio do not receive the lens covers they won't be able to record the morning news, and the Walpole High students are in need of this news to learn what is going on around the school. The Walpole High sports teams need the cameras to record game film so that they can watch it and get better. News broadcasters require crystal clear conditions when filming so the lens can't have any scratches or smudges. Photographers need clear images so that they can sell them and make a living.

Mr. O'Farrell - Head of TV production at high school

Students of TV program

Viewers of High School Film Festival/News

Mr. Connolly - Teachers apart of the film festival

Mr. Alan - Teachers apart of the film festival

Sports Teams getting filmed at their games

Problem Importance (A6)

Camera lens covers are important because the lenses themselves are highly susceptible to being damaged otherwise. Many different things damage the lens, such as scratches, water, and dust. When this happens, the image quality of the photos taken by the camera often decreases, and the damage to the lens is sometimes irreparable. (DPReview staff) For this reason, it's important to create a cover for the lens to prevent unwanted damage from occurring in the first place. This cover would shield the whole lens while it's not in use, while being easily removable to allow for the most optimal filming possible.



An example of the scratches that a camera lens can have, which reduces image quality.



The camera in the Walpole TV Studio.

Works Cited:

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